Constraint Based Problem

PROBLEM

Question 1: Hospital Doctor Scheduling using Backtracking Search

Problem Statement

A hospital needs to assign **three doctors (D1, D2, and D3)** to **three shifts (Morning, Afternoon, and Night)**. The assignment must satisfy the following constraints:

- D1 cannot work the Night shift.
- D2 must be assigned to a shift before D3 (Morning < Afternoon < Night).
- Only one doctor can be assigned to each shift.
- D3 cannot work the Morning shift.
- Every doctor must be assigned exactly one shift.

Apply the **Backtracking Search Algorithm** to determine a valid schedule. Show all intermediate assignments, perform constraint checking at each step, and state the final valid schedule.

Question 2: Robot Grid Navigation using Breadth-First Search (BFS)

Problem Statement

A robot operates in a **5 × 5 grid environment** and must travel from the **Start (S)** position to the **Goal (G)** position. Some cells contain obstacles that block movement.

The following constraints apply:

- The robot can move only **Up, Down, Left, or Right**.
- Each movement has a cost of **1**.
- The robot cannot pass through obstacle cells.
- Use the **Manhattan Distance heuristic** for reference.

Apply the **Breadth-First Search (BFS)** algorithm to explore the grid. Show the step-by-step node expansion, queue updates during traversal, determine the optimal path, and calculate the total cost.

Question 3: Autonomous Rescue Robot in a Disaster Environment

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a **partially observable environment**, where some paths are blocked and only neighbouring cells can be sensed.

The building is represented as a grid where the robot starts at **S** and must reach the survivor located at **G**.

The following constraints apply:

- The robot can move **Up, Down, Left, or Right**.
- Obstacle cells cannot be traversed.
- The robot can observe only adjacent cells.
- Each movement has a cost of **1**.
- Entering a risky zone incurs an additional cost of **2**.
- The robot must dynamically update its knowledge of the environment during navigation.

Develop an appropriate AI-based solution using **Uniform Cost Search (UCS)** or an **Online Search Agent**. Explain how intelligent agents, rationality, and environment type influence the robot's decision-making, and justify why the chosen search strategy is suitable.

Question 4: Fraud Detection in Financial Systems

Problem Statement

A financial institution processes thousands of transactions every day and requires an intelligent system to detect potentially fraudulent transactions. Each transaction contains information such as transaction amount, location, account number, transaction time, and transaction frequency.

The fraud detection system should analyse transaction patterns to identify suspicious activities based on predefined rules.

The objectives of the system are:

- Read transaction details from the user.
- Analyse each transaction using fraud detection rules.
- Identify suspicious or normal transactions.
- Display the classification result for every transaction.
- Improve financial security by detecting abnormal behaviour.

Develop an AI-based fraud detection system that analyses transactions and classifies them as **Fraudulent** or **Legitimate**.

Question 5: Convex Hull using Brute Force Algorithm

Problem Statement

A robotics system receives a set of obstacle coordinates and must determine the outer boundary enclosing all obstacles using the **Brute Force Convex Hull Algorithm**.

The tasks are:

- Explain how the Convex Hull of a set of points is determined.
- Implement the Brute Force Convex Hull algorithm.
- Analyse the time complexity of the algorithm.
- Discuss the scalability issues when the number of points increases.

The program should accept a set of two-dimensional coordinates as input, compute the Convex Hull, and display the boundary points forming the convex polygon.